Introduction to Java: Eclipse Setup Guide

Step-by-step instructions for installing JDK and Eclipse, creating a project, and using the built-in terminal. Note – the versions may vary – as long as your JDK works!

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1. Install the Java Development Kit (JDK)

Eclipse runs on the Java platform, so you need a **Java Development Kit (JDK)** before installing the IDE. A JDK **includes** the runtime (JRE) plus tools and source code.

⚠️ Make sure you download the **JDK (Java Development Kit)** and not just the **JRE**.

1. Download the JDK

Visit the Eclipse Temurin downloads page and select the long-term support installation. We recommend **JDK 21 (LTS)** for **Windows x64**. Choose the `.msi` installer for easiest installation).

Using **JDK 21 ensures compatibility with Eclipse and the course tools.**

2. Run the installer

After the download completes:

- Double-click the .msi file
- Accept the license agreement
- Use the **default installation path**

You usually do **not need to change any settings**.

3. Verify the installation

Open a Command Prompt (press `Win + R`, type `cmd`, press **Enter**) and run `java -version`. You should see the current version information.

💡 *Oracle vs Temurin distribution: If you need a completely free, production-ready JDK with multi-year LTS updates, Temurin is generally the go-to choice.*

If you require Oracle's commercial support guarantees (SLA), or depend on their specific certification and branding, Oracle JDK with a support subscription may be preferable.

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2. Install the Eclipse IDE

1. **Download the installer**: Go to the official Eclipse Installer download page and download the **Windows** x86_64 (**64-bit)** installer.

2. **Run the installer**: When the installer opens, select **"Eclipse IDE for Java Developers"** . This package includes everything you need for Java programming.

Figure 1. The Eclipse Installer lists available packages; choose "Eclipse IDE for Java Developers."

3. **Choose an installation folder**. The default location is usually fine. Click **Install** to begin. The installer will download and install the IDE automatically.

4. **Select your Eclipse workspace** 🏠

On first launch, Eclipse will ask for a **workspace directory**. The workspace is where Eclipse stores project files, settings, metadata, and preferences.

Choose a location (e.g., `C:\Users\YourName\eclipse-workspace`) and click **Launch**

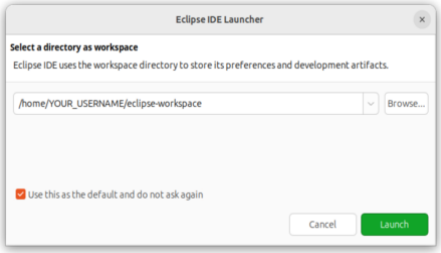


Figure 2. Eclipse prompts for a workspace folder on first launch.

🎯 *Tip: Check the "Use this as the default and do not ask again" box if you do not want Eclipse to prompt you every time it starts.*

Important for CSSE 220: Java 25 in Eclipse. For this course we recommend using JDK 21 (LTS) with Eclipse.

5. **Verify Eclipse sees your JDK**. Go to Window → Preferences → Java → Installed JREs. You should see your installed JDK 21 listed. If it is not listed, click Add → Standard VM and select your JDK installation folder. This prevents the "Eclipse cannot find Java" problem

3. Create a Java Project, Package, and Class

Time to create your first project! 😊

1. In Eclipse, from the top menu, go to **File** → **New** → **Java Project**.
2. The New Java Project wizard opens. Enter a project name**:** HelloWorld
3. **Leave Use default location checked**
4. Uncheck *Create module-info.java* (modules aren't needed and it will complicate things)!
5. Click **Finish**.

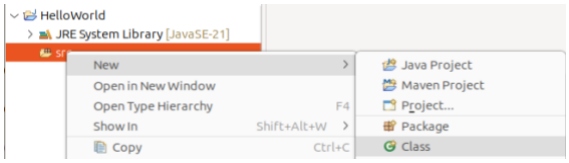
Your workspace folder contains all your projects.

Create a package

In the *Project Explorer* on the left, right-click the `src` folder and select **New** → **Package**. Enter a package name, such as `introduction (lower case)`, and click **Finish**. Packages organize classes.

Create a class

Right-click the package you just created and choose **New** → **Class**. Name the class (e.g., `HelloWorld`). Check the box to generate a `public static void main(String[] args)` method and click **Finish**.



When you click **Finish**, Eclipse will create your class and open it in the editor. Type:

```
System.out.println("Hello,world!");
```

inside the main method, save the file, and click the **Run** button (the green play icon) to see your program output in the console.

```
1 package introduction;
2
3 public class HelloWorld {
4     public static void main(String[] args) {
5         System.out.println("Hello, World");
6     }
7 }
```

Lost or stuck? 😕 No worries!

1. In Eclipse's top menu, click **Help** 📖 → Choose **Window** → **Tutorials** 🛠️
2. Select the **Hello World** tutorial and work through the steps 🧑‍🏫🧑‍🏫

3. Eclipse Shortcuts Every Undergrad Should Know

- **Indent:** `Ctrl + I`
- **De-Indent:** `Shift + Tab`

- **Toggle line comment:** `Ctrl + /`
- **Block comment:** `Ctrl + Shift + /` (wraps selection with /* */)

- **Move line up/down:** `Alt + ↑` / `Alt + ↓`
- **Duplicate line:** `Ctrl + Alt + ↓`

- **sysout snippet:** type `sysout` → press `Ctrl + Space`
- **General auto-suggest:** `Ctrl + Space`

- **Maximize/restore current panel:** `Ctrl + M`

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5. You're Ready to Code!

Congratulations 🎉 – you've set up your Java development environment! You installed the JDK and Eclipse, created your first project, package and class. Have fun and keep learning! 🚀